AI guided discovery of cellular rejuvenation mechanisms

Technical white paper and venture development plan

David Xu, PhD Peter Lee, PhD, MD Zhimin Zheng, PhD

20 October 2026

We propose a research company that uses artificial intelligence and prospective experiments to discover how adult cells can recover youthful function while retaining their identity. Our long-term objective is comprehensive reversal of cellular biological aging. Our first financing would establish whether a small, controllable intervention can cause durable functional rejuvenation in a defined human tissue model.

The investment thesis is that the scarce asset in rejuvenation research will be reliable causal knowledge: which intervention works, in which cells, for how long, with which risks. We will build a longitudinal dataset linking interventions to molecular state, tissue function, identity and safety, and use it to improve the selection of subsequent experiments. Models will earn their place by increasing independently confirmed discoveries per dollar.

The proposed initial program is a 24-month, $8.0 million discovery and validation plan. It focuses on aged human skin cells and reconstructed skin, with impaired tissue repair as a provisional disease entry point. This financing would support target discovery and lead feasibility. Clinical development and a whole-body rejuvenation program would require additional capital and evidence.

Current evidence consists of an earlier research framework with public-data and synthetic computational pilots. No proprietary rejuvenation target, biological efficacy result, clinical asset or human age-resetting treatment is claimed. The opportunity is a disciplined attempt to create those assets.

Investment decision

Fund the program if the founding team can secure clean intellectual property rights, experienced experimental leadership and a reproducible assay that distinguishes restored function from proliferation and cell selection. Release later spending against prospective evidence gates. Stop or narrow the program if causal effects or an acceptable therapeutic window fail to replicate.

Nonconfidential research and business discussion. Budget, pricing and commercial milestones are planning assumptions. This document contains no securities terms or offer to sell securities.

Investment proposition and present evidence

Our initial product is a validated rejuvenation mechanism and an intervention package suitable for pharmaceutical partnering. The enabling platform combines multimodal data analysis, causal modeling, experimental design and reproducible laboratory execution. Broad cellular age resetting remains the scientific ambition; the financing case rests on one measurable tissue program.

Investor question	Present position	Value created by the seed program
What has been demonstrated	A treatment-expression signature in public data and falsification tests in simulations	Independent biological confirmation of intervention effects
What is proprietary today	No proprietary biological asset is represented	Rights-cleared longitudinal perturbation data and specific inventions
What is being financed	Discovery infrastructure and one tissue program	A reproducible mechanism package and delivery feasibility
What remains uncertain	Causal targets, durability, safety, generalization and clinical benefit	Experiments that reduce or expose these uncertainties

The earlier 13 October 2026 manuscript reports 34 recovery samples from three donors in public RNA-sequencing data. Donor-holdout treatment classification achieved AUROC 1.00. This establishes that the selected treatment groups can be separated in that dataset. Response-based sorting, induction-related expression and the small number of independent donors limit interpretation. The result does not measure reversal of aging. The paper also reports six synthetic settings over 20 seeds each, showing how clock optimization can select an ineffective reporter and how missing combination data defeats mechanistic identification. [1]

These pilots support the design of an experiment-selection system. They provide no estimate of therapeutic success probability, clinical effect size or human lifespan extension. Their reported reproducibility is an engineering starting point; an investor diligence replay should verify the archived code, data manifest and exact outputs before reliance.

The first three months therefore purchase assay qualification, data rights review, computational replay and experimental leadership. Those deliverables are essential assets even if the initial hypothesis fails. The company should avoid a valuation narrative based on a perfect classification score or on rejuvenation claims made by other companies.

Our proposed differentiation is prospective confirmation under a fixed experimental budget, with explicit alternative explanations and a record of negative results. This differentiation must be demonstrated against expert selection and simple statistical models.

Scientific basis and the meaning of a complete reset

Embryonic development and induced reprogramming suggest that some age-associated cellular states can be reversed. Analyses of embryonic methylomes identify declines in clock estimates during development, and transient reprogramming studies report molecular and selected functional changes in adult cells. These observations motivate a search for causal control mechanisms. They do not establish one universal aging switch or erasure of every form of damage. [2–5]

Cellular aging includes interacting processes such as genomic instability, epigenetic change, mitochondrial dysfunction, impaired proteostasis, senescence and changes in intercellular communication. Some damage may require physical repair, removal or replacement. A regulator that changes gene expression cannot be assumed to correct acquired DNA mutations, extracellular crosslinks or tissue architecture. [6]

We define rejuvenation operationally as movement toward an independently established youthful functional reference, with preserved lineage identity and acceptable integrity. A clock is a measurement model trained against a particular target. Its numerical output is one component of evidence and may become unreliable in embryos, diseased tissue or strongly perturbed cells.

Dimension	Required evidence	Interpretation boundary
Molecular state	Concordance across frozen methylation and expression measures	A lower score alone cannot establish benefit
Function	Improvement in a prespecified tissue-relevant assay	General stimulation can improve an assay without reversing aging
Identity	Retained lineage markers and differentiated function	Dedifferentiation can mimic youth-associated profiles
Integrity	Genome and growth surveillance appropriate to the intervention	Short observations cannot exclude delayed cancer risk
Persistence	Benefit after withdrawal and under a later challenge	A transient response cannot establish a stable reset
Breadth	Independent confirmation across contexts and tissues	One cell type cannot establish a universal mechanism

Complete cellular age resetting is a long-term hypothesis about coordinated restoration across these dimensions. There is currently no universally accepted assay that proves such completeness. We will state exactly which dimensions, tissues and time horizons have been tested. Whole-organism rejuvenation would require additional evaluation of immunity, extracellular damage, organ coordination, delivery and long-term disease outcomes.

The research program remains valuable if control is distributed across several mechanisms. A reproducible, bounded intervention with meaningful function can become an asset without establishing a universal clock.

Competing mechanistic hypotheses

We will test four explanations in parallel. This prevents the model and laboratory program from assuming that the desired master switch already exists. Developmental observations guide hypotheses; adult-cell interventions establish whether they are actionable.

Hypothesis	Distinguishing observation	Consequence for development
Concentrated regulatory control	A small intervention set reproduces several independent benefits and passes rescue tests	Optimize a bounded control program
Distributed maintenance network	Several partially substitutable mechanisms are needed across contexts	Develop combinations with explicit interaction data
Repair and replacement constraints	Molecular changes occur while damage or tissue function remains impaired	Add a repair modality or discontinue the candidate
Measurement or population artifact	Apparent youth disappears after composition, division and survival controls	Reject the mechanistic claim

The initial search includes transcriptional regulators, chromatin maintenance, proteostasis, mitochondrial quality control, senescence-associated signaling and extracellular communication. These are discovery categories. No named target is represented as a newly discovered mechanism. Developmental regulators require special caution because activities that are useful in an embryo may destabilize an adult cell.

Each candidate receives an evidence record containing the intervention definition, biological context, temporal sequence, independent reagents, functional effect, uncertainty and known counterexamples. Association and model attention remain hypothesis-generating evidence. A claim advances only when an experiment changes the candidate mechanism and measures the predicted downstream outcome.

Mechanistic confirmation has six parts. Temporal precedence places the candidate change before the outcome. Necessity tests whether disrupting it weakens a reference response. Sufficiency asks whether the targeted change produces part of the benefit independently. Rescue restores the effect using an orthogonal approach. Withdrawal and challenge experiments measure persistence. An independent laboratory tests reproducibility. A necessity result is interpreted relative to perturbation toxicity and target engagement.

We will distinguish increased repair from reduced damage production, dilution through division and preferential survival of a youthful subpopulation. Division tracking, absolute cell counts and lineage or population tracking are required wherever feasible. A finding confined to surviving cells will be reported with the full survival distribution.

Failure of the sparse-control hypothesis is scientifically informative. It changes the intervention strategy and capital requirement; it should not trigger indefinite searching for a more convenient clock score.

Data strategy and commercial rights

Public resources can support representation learning, assay selection and reproducible baselines. Proprietary value must come from prospective, rights-cleared intervention data with functional follow-up. We will preserve study, donor and specimen identity so that large cell counts cannot disguise a small number of biological replicates.

Resource	Proposed use	Critical limitation
GSE165180 with GSE165177 and GSE165179	Human transient-reprogramming expression and methylation time courses	Modalities and cultures require explicit matching [7]
GSE113957	Adult human fibroblast age references	Cross-study culture and donor effects
GSE49828	Human developmental methylation comparisons	Development differs from adult maintenance
GSE121708 and linked subseries	Mouse gastrulation multiomics	Species and developmental transfer
GSE197437	Partial-reprogramming response comparisons	Cell-type and induction dependence
GSE132042	Adult mouse tissue-aging references	Animal evidence cannot establish human efficacy
GSE133344 and genome-scale Perturb-seq	Regulatory priors and interaction baselines	Transformed cells have limited relevance to normal aging [8]
Prospective adult donor studies	Function, withdrawal, identity and safety under intervention	Consent, cost and reproducibility govern scale

The catalogue above extends the resources described in the earlier research framework. Only the GSE165177 subset was analyzed in its public-data pilot. New acquisitions will be checked against original accession records before use. A public accession provides discoverability, not an automatic commercial license or evidence that modalities are paired. [1,7]

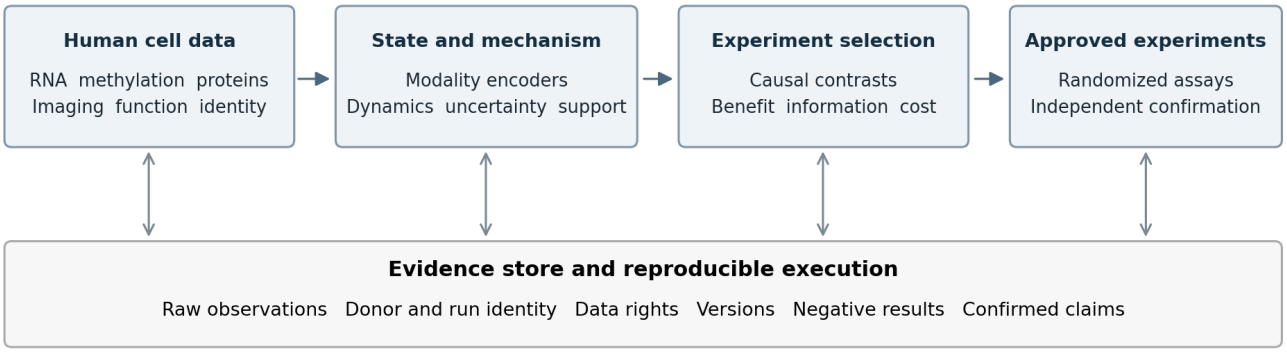
Every dataset requires a manifest with source and checksum, species, assembly, assay, donor, age, sex where permitted, condition, exposure, withdrawal interval, passage, batch and missingness. Regulatory elements retain genomic coordinates and the basis of gene linkage. The system will preserve unprocessed data and record transformations as versioned outputs.

A commercial rights register will cover consent, permitted uses, redistribution, geographic restrictions, model training, derivative data and partner obligations. Human genomic and health data receive access controls and a lawful transfer assessment for each jurisdiction. Restricted data will remain in approved environments. Federated training is an option when necessary; it cannot cure defective consent or incompatible rights.

The durable dataset is a matched record of action, time, molecular response and function, including failures. Data volume without that structure has limited value for identifying a rejuvenation mechanism.

Technical architecture

The platform will connect four computational layers to an experimental evidence store. Existing models will be adapted before funding a new foundation model. Proposed architectures are design choices to benchmark; they are not descriptions of a trained proprietary system.



Proposed architecture. Model predictions return to experiment selection through the evidence store.

Modality encoders will process RNA counts, methylation, accessibility, proteins and imaging. Actual paired measurements support cross-modal training. Unpaired assays use explicit missingness masks, dataset identifiers and uncertainty. The system will not invent a matched protein or methylation measurement from an RNA sample and treat the prediction as ground truth.

A regulatory representation links transcription factors, regulatory elements, genes and measured phenotypes. Sequence models can nominate regulatory regions, while perturbation models predict responses to actions. scGPT, GEARS and AlphaGenome provide examples of relevant model classes. Their use would depend on licensing, assay compatibility and prospective performance. [8–10]

An action-conditioned dynamics model receives the present state, target, direction, delivery class, exposure, withdrawal time and biological context. Separate output heads estimate function, identity, viability and integrity signals, along with uncertainty and evidence support. A useful first implementation is a regularized interaction model beside a modest graph or latent-state model. Larger architectures are justified only by held-out improvement.

The planner proposes informative, feasible experiments. A laboratory information system records materials, randomized assignments, instrument outputs and deviations. Acceptance requires predefined criteria and independent review. The evidence store links every claim to raw data, analysis version and a reproducible calculation.

Language models support literature extraction, code generation, protocol comparison and error diagnosis. They cannot certify a causal mechanism or authorize a new biological procedure. Approved protocols and laboratory personnel control execution. AI agents can use a general research orchestration layer, including Future-Code where rights and validation permit; no productivity or biological-performance claim is assumed for that integration.

AI methods for discovery and measurement

We will maintain a broad technology portfolio and allocate experiments according to demonstrated benefit. Each method has a specific output and a benchmark that determines whether it remains in the program.

Method	Scientific output	Adoption test
Retrieval and language models	Source-linked hypotheses and contradiction maps	Citation accuracy, expert review and useful novel hypotheses
Single-cell and multimodal models	Representations of cell state and response	Improvement over PCA, regularized models and frozen features
Sequence regulatory models	Candidate cis-regulatory elements and context predictions	Targeted experimental validation beyond sequence scores
Graph learning	Context-dependent regulatory interactions	Better prediction on held-out targets and combinations
Causal inference	Explicit estimands and alternative explanations	Prospective perturbations, rescue and sensitivity analyses
Dynamic state models	Predicted induction and withdrawal trajectories	Calibration on independently observed time courses
Computer vision	Cell morphology, counts, identity and functional image readouts	Blinded annotation, instrument transfer and artifact checks

The literature does not justify assuming that a foundation model will outperform a simple predictor. Published benchmarks show that results depend on strong baselines, systematic variation and the metric used. A recent response also reports settings in which deep perturbation models outperform baselines under calibrated metrics. We will resolve this question in our own prospective setting using shared data, endpoints and budgets. [11–13]

Model evaluation will use donor, laboratory, study, intervention and time holdouts. All cultures and derived measurements from one donor stay together. Feature selection and normalization parameters are fitted within training data. Pretraining overlap and study provenance will be documented where knowable. A study that contributed to pretraining cannot automatically serve as an independent generalization test.

We will score perturbation-induced changes relative to matched controls, performance on affected genes selected without test-label leakage, and independent functional outcomes. A high correlation across largely unchanged genes is insufficient. Models will also be judged on calibration, abstention and the number of experiments needed to confirm an effect.

No method advances because it is fashionable. Each model has a recorded cost, version, supported domain and retirement criterion. A simpler model that yields the same confirmed discoveries at lower total cost remains the preferred tool.

AI methods for intervention design and execution

The next group of methods becomes useful once assays and intervention data are reliable. Broad technical coverage does not imply simultaneous spending on every approach.

Method	Intended use	Boundary and timing
Bayesian optimization and active learning	Select informative targets and combinations under a fixed budget	Begin with conservative uncertainty and explicit exploration
Experimental design and interaction models	Separate main effects, synergy and redundancy	Measure relevant combinations rather than infer them from single actions
Generative chemistry	Propose compounds after a tractable mechanism is established	Require assay activity, selectivity and developability
Protein structure and design	Explore binders or delivery components when the modality needs them	A plausible structure cannot establish cellular efficacy [14]
Offline reinforcement learning and control	Evaluate multistage intervention schedules	Defer until action support and off-policy validation are adequate
Robotics and laboratory scheduling	Execute approved repeatable workflows and reduce waiting	Human-approved procedures, instrument QC and traceable exceptions
Federated and privacy-preserving learning	Analyze restricted partner data across environments	Use only when governance requires it and accuracy remains adequate
Research agents and formal software checks	Track claims, runs, costs and reproducible analyses	Verify outputs and preserve recoverable experiment state

A portfolio review every quarter will compare the marginal cost of another model with the value of another experiment. The first-year compute envelope is deliberately smaller than laboratory and personnel spending. We will adapt pretrained systems and maintain a compact model ensemble rather than start with frontier-scale pretraining.

For experiment selection, the planner estimates functional benefit and information gain, subtracts assay and execution cost, and rejects actions that exceed identity, viability or uncertainty limits. These terms and their weights are frozen for a prospective comparison. Diversity requirements preserve exploration, including experiments likely to disprove the current hypothesis.

Unsupported combinations receive an insufficient-evidence label. No amount of confidence calibration can identify an interaction absent from the action support without additional assumptions. If two mechanisms make the same predictions on observed interventions, the next experiment should distinguish them. The earlier synthetic study illustrates precisely this failure mode. [1]

Quantum machine learning, unrestricted autonomous agents and new general foundation-model training are optional research directions. They receive funding only after a concrete task and measurable incremental value are established. The central scarce resource remains independent biological evidence.

Model specification and reproducible evaluation

The first implementation will train on intervention-level and donor-level units. A record contains specimen identity, baseline measurements, action, measured target engagement, observation time, outcome, assay batch and data rights. The output includes predicted changes in each endpoint, a prediction interval, applicability flags and source experiments.

We will begin with no-change, mean-response, ridge-regression and regularized interaction baselines. Candidate advanced models include a graph encoder and a low-dimensional stochastic state model. Dimensionality, regularization and ensemble size will be selected inside nested training folds. Modality-specific likelihoods accommodate counts, methylation coverage and continuous imaging features. Missing modalities are masked in the objective, and predicted modalities remain labeled predictions.

The primary causal quantity is the difference in expected functional outcome between the assigned intervention and its matched control in a stated donor population and time window. Randomization identifies that contrast under the protocol assumptions. Observational adjustment provides supporting analyses and cannot replace the prospective intervention comparison.

The model-selection objective combines prediction error with calibration and context-transfer performance. The biological decision uses separate constraints for identity, viability and integrity. We will avoid a single composite score that lets a favorable clock compensate for harmful growth or loss of identity. A hard safety exclusion remains a hard exclusion regardless of the predicted benefit.

Evaluation level	Unit and split	Principal measure
Molecular prediction	Donor and unseen intervention	Error on intervention-induced changes
Function prediction	Donor and laboratory	Error, calibration and direction of effect
Experiment selection	Fixed-budget candidate arms	Independently confirmed hits per dollar
Mechanism	Orthogonal reagents and rescue	Concordance with prespecified causal predictions
Reproducibility	External laboratory and bridge specimens	Effect direction, magnitude and uncertainty

Every reported result will include the independent sample size, exclusions, uncertainty and prespecified analysis. Container images, locked dependencies, data hashes, seeds and experiment identifiers support replay. Human-readable claim records distinguish proposed experiments from executed runs and synthetic tests from biological measurements.

An agent failure, instrument interruption or model outage must preserve experiment identity and raw observations. The system may retry an idempotent analysis. It must not silently repeat a biological exposure or treat an incomplete laboratory run as a completed negative result.

Mechanistic dynamics and the path to broader reset

We will test whether a bounded regulatory intervention changes the rates at which measurable damage is produced and removed. A deliberately simple candidate model relates a measured damage burden D to regulatory activity r and biological context c.

$$\frac{dD}{dt} = k_{\mathrm{in}}(c) - k_{out}(r,c)\ D$$

$$\frac{dr}{dt} = a(u,c) + b(r,c) - \gamma r$$

Here the input term denotes damage production, the output coefficient denotes the removal rate, u is the intervention and the Greek letter gamma denotes a relaxation rate. The function b represents endogenous feedback. These equations are testable candidate descriptions, not an established law of aging. Each damage variable needs an independent assay; an arbitrary latent coordinate cannot supply its biological interpretation.

Pulse, withdrawal and controlled-challenge observations help separate altered production from enhanced removal. Division tracking and absolute counts test dilution and cell selection. Competing models will be compared on held-out intervention trajectories and recovery times. A stable-looking fitted equilibrium is insufficient unless perturbation and recovery measurements support it. Identifiability will be assessed before interpreting a fitted parameter as a causal repair rate.

This framework makes a deeper mechanism claim falsifiable. A regulator controlling coordinated restoration should alter multiple independently measured processes in the predicted temporal order. Perturbing that regulator should weaken the response, and an orthogonal rescue should restore it. If several networks explain the same observations, experiments should distinguish their predictions rather than select the most attractive story.

Expansion follows evidence. The first stage establishes control in one adult tissue. A second stage tests a distinct lineage with an independently calibrated functional reference and delivery approach. Later stages examine tissue interactions, immune effects, extracellular damage and organism-level function. Acquired mutations and other residual damage may require separate repair or replacement strategies.

Each expansion requires a new budget and specific safety studies. The initial financing does not fund these later stages. A complete reset would require coordinated restoration across measured aging dimensions and acceptable long-term organismal outcomes. Failure in any essential dimension limits the claim, even when another clock or tissue improves.

Prospective discovery and confirmation design

The initial tissue program uses primary adult dermal fibroblasts and later fibroblast–keratinocyte reconstructed skin. Impaired repair is a provisional development direction. An intervention affecting a fibroblast assay may still fail in a wound because vascular disease, inflammation and tissue architecture influence healing.

Discovery will compare 180 distinct candidate targets across eight adult donors using three independent targeting reagents and two technical assay wells per condition. This gives 8,640 candidate donor reagent wells. A 20% planning allowance yields about 10,368 wells including controls. Wells improve assay precision; the independent biological sample size remains eight donors. Young-reference specimens are a separate reference set. Target engagement and delivery controls are included.

Candidate selection allocates 60 targets to a locked model, 60 to blinded expert nomination and 60 to a prespecified random eligible set. Duplicates are resolved before experimentation by a recorded replacement rule. All arms use the same eligibility filters, reagents, donor allocation, plate randomization and confirmation criteria. The comparison tests the complete selection method at equal cost.

Tier one measures assay-qualified cellular phenotypes and excludes toxic or identity-disrupting actions. These are screening signals. Up to 12 candidates advance to functional confirmation in reconstructed skin, with up to four prespecified slots per selection arm. The primary endpoint is chosen and locked during assay qualification. Candidate endpoints include recovery of barrier function or re-epithelialization after a standardized injury, with cell number and proliferation analyzed separately. Clocks remain secondary.

Confirmation uses 16 new adult donors, assigned across two laboratories, with a subset of four bridge donors measured at both sites. The final allocation will balance relevant donor characteristics and disease status where feasible. Necessity, sufficiency, rescue and withdrawal studies focus on the strongest mechanisms. Selected candidates receive orthogonal molecular profiling; comprehensive single-cell multiomics is reserved for informative subsets.

The budget includes planning volumes, not vendor quotes. A roughly $0.95 million primary-screen allocation corresponds to about $92 per planned assay well, including controls, before separately budgeted central personnel. This assumption must be replaced with laboratory quotations in the first quarter. If the qualified assay costs substantially more, target count or program scope will be reduced before committing the full screen.

Evidence gates and statistical decision rules

The program begins with assay qualification. Acceptance thresholds below are proposed development criteria; they are not observations or universal definitions of rejuvenation. Their final values will be locked using pilot variance, biological relevance and independent scientific review before testing confirmation candidates.

Timing	Required deliverable	Spending decision
Months 0 to 3	Clean rights register, experienced experimental lead, replayed pilots and qualified control assays	Release the discovery screen only after review
Months 4 to 6	Balanced candidate screen, usable target engagement and complete failure records	Advance up to 12 candidates within the original budget
Months 7 to 12	At least two candidate mechanisms with concordant orthogonal evidence and functional benefit	Select one program or narrow the platform
Months 13 to 18	New-donor and external-laboratory replication with withdrawal follow-up	Fund delivery and lead-feasibility work if effects persist
Months 19 to 24	One mechanism package with a feasible modality, IP assessment and next-study plan	Seek a partner or further financing; discontinue failures

A useful proposed functional hurdle is recovery of at least 20% of the prespecified old-to-young functional gap in the confirmation assay. It applies only if that gap is reproducible and the assay has adequate precision. If no meaningful reference gap exists, the assay is unsuitable. Absolute effect sizes and clinical relevance must accompany this normalized measure.

Confirmation also requires a prespecified confidence interval excluding no improvement on the primary endpoint, consistent direction across donor analyses, retained identity within a biologically justified noninferiority margin, acceptable viability and no disqualifying integrity or abnormal-growth signal. Multiple candidates and endpoints require a locked multiplicity strategy. Exploratory screens may control false discovery rate; confirmatory claims require their own prespecified error control.

Withdrawal observations at roughly 4 and 12 weeks are proposed research milestones. Exact duration depends on the qualified tissue model and may require a separate longitudinal system. Longer safety assessment follows the modality. These windows cannot establish lifelong persistence or absence of delayed malignancy.

Power will be calculated from donor-level variance during assay qualification, including clustering, dropout and the primary contrast. Sixteen confirmation donors are an initial budget assumption. If power is inadequate, the company must narrow the claim, increase the cohort with explicit funding, or stop. Technical replicates cannot repair inadequate biological replication.

Safety and the limits of biological reversibility

Safety is part of the definition of a useful rejuvenation intervention. A cell that appears young while losing lineage control or expanding abnormally fails the program. Developmental and reprogramming pathways can alter proliferation and identity; apparently favorable molecular signatures therefore require a separate assessment of these outcomes.

We will monitor identity and differentiated function, target engagement, viability, population composition, abnormal growth, genomic integrity and relevant stress responses. Assay selection follows the intervention. Screening readouts cannot exclude every oncogenic or mutational risk, so a lead program requires a risk-based follow-up plan with appropriate independent expertise.

Failure mode	Early detection	Development response
Dedifferentiation	Identity and function panels over induction and recovery	Reject the candidate or reduce the intervention scope
Expansion of abnormal clones	Absolute counts, population tracking and growth surveillance	Stop advancement and investigate mechanism
Clock-only improvement	Independent function and damage measurements	Reject the rejuvenation claim
Selection of resilient survivors	Survival distributions and lineage or population tracking	Distinguish selection from restoration
Rapid rebound after withdrawal	Serial function and molecular observations	Reassess mechanism and required treatment burden
Delivery or systemic toxicity	Modality-specific distribution and tolerability studies	Change modality or discontinue

The preferred initial modality will be chosen after mechanism confirmation. Transient, controllable exposure may offer practical advantages, but it does not inherently establish safety. Small molecules, RNA, gene delivery and ex vivo approaches have different exposure, manufacturability and long-term risk profiles. Target feasibility and freedom to operate influence the choice.

Genome mutations, mitochondrial heterogeneity, extracellular matrix damage and immune aging may persist even after a favorable epigenetic response. A single intervention may therefore have a limited scope. The platform will record residual damage and explicitly test whether adding another modality provides additional functional benefit.

The first 24 months fund preclinical discovery and lead feasibility. No self-experimentation, clinical administration or treatment recommendation is proposed. Human translation requires an appropriate sponsor, manufacturing controls, authorization, ethics review and a protocol justified by the resulting evidence. Animal work, if later required, must follow an approved, independently reviewed plan with relevant endpoints and adequate follow-up.

Regulatory development and data governance

The initial clinical strategy, if a lead emerges, will target a defined disease and a meaningful functional outcome. A general claim of reversing aging across the body would require much broader evidence. The exact regulatory route depends on the product, indication and jurisdiction and will be discussed with the relevant agency as the program matures.

We will treat epigenetic and transcriptomic clocks as exploratory or pharmacodynamic biomarkers unless a specific context of use is supported and accepted. FDA resources distinguish biomarkers from clinical outcomes and require substantial evidence before a surrogate can replace a clinical outcome. A device measuring a biomarker and a qualified surrogate endpoint are different regulatory matters. [15,16]

For a chronic-wound direction, the program must eventually justify clinically relevant healing outcomes, durability, infection and recurrence assessment, and an appropriate comparator reflecting standard care. The preclinical model provides mechanistic and functional support. It cannot establish patient benefit or define the clinical endpoint by itself. The provisional indication will be reconsidered after delivery and tissue-model evidence.

Gene therapy can require surveillance for delayed adverse events, with follow-up duration based on product characteristics and risk. The initial seed budget does not include a clinical gene-therapy development package. FDA guidance on long-term follow-up informs the later plan; it does not remove the need for product-specific consultation. [17]

The company will maintain research consent and tissue-procurement records, donor privacy controls, access logs and a commercial-use rights register. Cross-border collaboration requires specific review of human genetic resources, personal data transfers, contractual restrictions and institutional obligations. A geographic expansion plan must follow the actual data and product footprint.

Software will separate source data, derived measurements, model predictions and accepted claims. Versioned analysis specifications, access control and audit trails will support diligence. A future regulated system would need validation appropriate to its intended use; ordinary research reproducibility does not automatically meet a clinical quality-system requirement.

We will seek regulatory and legal advice at defined decision points. The business case assumes no special exemption for longevity, no automatic acceptance of a clock endpoint and no shortcut from mechanistic plausibility to marketing authorization. Regulatory cost and timing remain material uncertainties until the modality and disease program are fixed.

Competitive position as of October 2026

The field already contains substantial scientific capability and capital. Our differentiation must survive comparison with those companies. The table describes publicly stated activities and our proposed positioning; it does not imply access to competitors' confidential data.

Organization or approach	Public position	Implication for this program
Life Biosciences	ER-100 partial epigenetic reprogramming in an optic-neuropathy Phase 1 study	Human translation is being explored; eye results cannot validate our tissue or mechanism [18]
NewLimit	Epigenetic reprogramming medicines; announced a $435 million Series C in June 2026	Capital and execution expectations are high; seek a narrow evidence advantage [19]
Altos Labs	Cellular rejuvenation programming and restoration of cell health	Competing on broad ambition alone offers little differentiation [20]
Retro Biosciences	Programs addressing cellular reprogramming and other aging mechanisms	Multiple mechanisms may be needed; scope and capital discipline matter [21]
General AI biology platforms	Cell representations, perturbation prediction and drug discovery	Commodity models weaken a software-only moat [8–14]
Existing disease treatments	Indication-specific clinical benefit and established development paths	A lead must eventually outperform meaningful clinical alternatives

On 8 October 2026, Life Biosciences reported interim observations in three glaucoma participants through day 56. The company described tolerability and preliminary visual-field signals in two participants. This small, uncontrolled interim company report is a development milestone. It cannot establish efficacy, comprehensive rejuvenation or long-term safety. Our plan does not extrapolate a human treatment claim from it. [18]

We propose to compete through a well-defined adult tissue context, prospective comparison of experiment selection, mechanistic falsification, independent functional replication and tractable intervention design. None of these capabilities is assumed to be unique. The defensible asset would be the specific validated mechanism, its intervention rights, and a dataset that helps predict when the effect fails.

Partner conflicts are a real constraint. Investors with competing portfolio companies may still be appropriate contacts, but detailed unpublished targets should be shared only after a conflict assessment and an appropriate confidentiality decision. Nonconfidential outreach should describe the program, evidence stage and financing objective without disclosing patent-enabling intervention details.

Product and commercialization strategy

The proposed company will develop one lead mechanism program and a reusable discovery platform. Its first commercial route is a research collaboration or option agreement with a pharmaceutical or biotechnology partner. Asset licensing becomes credible only after specific biological evidence and a rights assessment. A clinical product is a later possibility requiring further capital.

Offering	Buyer and decision maker	Required evidence
Defined discovery collaboration	Pharma discovery or external-innovation team	Reproducible assay, data rights and agreed deliverables
Mechanism or intervention option	Therapeutic-area or business-development team	Confirmation, target engagement, delivery feasibility and IP position
Asset license or co-development	Biotech or pharma sponsor	Partner-specific diligence and a funded development path
Research software or data access	Research teams with compatible needs	Repeated utility and clear rights; secondary to the lead program

An initial collaboration would specify the tissue, intervention scope, sample ownership, analyses, success criteria, publication review and foreground IP. Payment should track defined work and acceptance. The platform should retain rights to general methods and nonexclusive learnings where contractually possible. Exclusivity must be limited to a funded target, indication or program scope.

We will avoid depending on consumer longevity subscriptions, unvalidated age tests or direct clinical claims to finance discovery. A fee-for-service model can produce revenue but may divert the same laboratory team from proprietary work. Initial partnering will be capped at two concurrent programs unless separate resources are funded.

The first business-development work is customer discovery. We propose 20 structured interviews with pharmaceutical and biotech decision makers, followed by five technical evaluations and two negotiated pilot scopes. These are commercial targets. No customer, partnership or signed letter of intent is represented as secured.

A proposed paid pilot price of $0.5 million reflects an illustrative $0.3 million direct execution cost and $0.2 million contribution before platform overhead. The model requires a buyer willing to pay for the agreed evidence package. Price, margin, sales cycle and IP concessions must be validated through actual negotiations. A successful technical pilot without a credible buyer will trigger a commercial strategy review.

Market scope and the route to revenue

The population affected by aging is large, but population size does not establish the obtainable market for an early discovery company. We will size the initial opportunity using identifiable research buyers and realistic program capacity. Disease-product economics will be assessed after an indication and modality are selected.

For planning, a reachable list of 40 relevant pharma and biotech groups, an average of one eligible program per buyer and $0.5 million per program implies a $20 million annual collaboration opportunity. These are explicit assumptions, not a measured market estimate. The company must validate buyer eligibility, available budgets and procurement timelines through interviews.

With capacity for two concurrent pilots and an assumed two pilots completed per year, initial revenue capacity is $1.0 million annually at the illustrative price. This is substantially below the seed operating budget. Collaborations can validate demand and offset selected costs; they do not eliminate financing dependence. Therapeutic licensing economics are excluded from the base funding case because no licensable asset exists today.

Period	Planned activity	Revenue treatment
First 12 months	Assay qualification, discovery and buyer interviews	Base financing case assumes zero receipts
Months 13 to 24	Seek up to two paid collaboration pilots	Upside only until signed and collected
Years 3 to 5	Partner a validated mechanism or expand focused collaborations	No royalty or milestone forecast without an asset and contract

The provisional skin-repair direction will be tested against alternatives using measurable functional benefit, delivery accessibility, commercial unmet need, development cost and competition. A different tissue may become preferable, but a change requires a documented rationale and a revised budget. The seed program cannot expand to several organs without trading away depth or increasing capital.

Potential partner or acquirer categories include pharmaceutical groups with tissue-repair programs, companies pursuing reprogramming interventions and discovery platforms needing causal human-cell datasets. These are logical categories, not predicted transactions or verified interest. Exit value and timing cannot be inferred from competitors' financing valuations.

The commercial milestone at month 24 is a partner-diligence package and evidence of a buyer's willingness to engage under specific terms. A financing round or a collaboration may still fail despite sound science. Cash planning therefore assumes no uncontracted revenue and includes a decision to narrow or wind down the program if the next financing is unavailable.

Capital plan and operating assumptions

We propose an $8.0 million seed plan covering 24 months of discovery and lead feasibility. The amount is an illustrative financing requirement, subject to actual hiring plans, laboratory quotations and company formation. It is not a committed round or a statement of existing cash. Clinical trials, GMP manufacturing and a full IND-enabling package are outside this budget.

Use of funds	USD millions	Scope and assumption
Personnel and scientific leadership	3.20	Approximately 10 average paid FTE equivalents at $160,000 annual fully loaded cost over two years
Laboratory materials and assay operations	2.10	$0.95m primary screening, $0.40m omics, $0.25m tissue models, $0.20m validation materials, $0.30m specimens and QC
Compute and research software	0.60	Hosted compute, storage, model access and reproducible analysis
External replication and specialist studies	0.80	Independent-lab work and specialist assessments with distinct scopes
IP legal and regulatory preparation	0.35	Formation, ownership review, filings and initial advice
Operating overhead	0.35	Shared space and administration outside directly allocated assay charges
Uncommitted contingency	0.60	Released only against a reviewed variance or critical experiment
Total proposed financing	8.00	$7.40m operating plan plus $0.60m reserve

Laboratory operations exclude central personnel already counted in payroll. External replication covers separately contracted work rather than duplicating internal assay charges. The $160,000 blended cost assumes a mixed seniority and geography team with contracted support; a predominantly senior US team could cost materially more. Vendor quotations and an approved hiring schedule replace these assumptions before major commitments.

Planned quarterly operating outflows are $0.55m, $0.70m, $0.86m, $0.95m, $1.03m, $1.08m, $1.10m and $1.13m, totaling $7.40m. With the full financing at the start and no receipts, ending cash is $0.60m. Average operating burn is about $308,000 per month, but late-program spending is higher than the average.

A 30% increase in laboratory and external-study costs adds $0.87m, exceeding the reserve by $0.27m. The company would need to reduce scope, delay work or raise more capital. Reserve cash is not a claim of guaranteed runway. A quarterly cash review will compare actual commitments, completed evidence and remaining funding, and preserve an orderly-stop budget before accepting new experiments.

Team ownership and intellectual property

The authors of this white paper are David Xu, PhD; Peter Lee, PhD, MD; and Zhimin Zheng, PhD. Formal roles, time commitments, equity allocation and relevant experience should be documented before financing. No employer or institution is represented as a sponsor of this venture.

David Xu's mathematics and AI work provides a computational starting point. A research company pursuing rejuvenation also needs an experimental leader with primary-cell and tissue-model experience, a translational development lead and independent expertise in aging, oncology risk and statistical design. The capital plan includes these functions; appointments remain part of company formation.

Function	First-year accountability	Required evidence of readiness
Computational leadership	Models, reproducibility and experiment selection	Replayed analyses and prospective benchmark
Experimental leadership	Qualified assays and interpretable perturbations	Relevant execution history and approved study plan
Translational leadership	Modality and indication decisions	Product development experience and risk assessment
Independent scientific review	Evaluate confirmation and stop decisions	Conflicts disclosed and raw-data access
Business and finance	Formation, budgets and partner negotiations	Authority, ownership records and controlled commitments

Before using any existing code, data or invention, the company will establish ownership and commercial-use rights. This includes prior employment obligations, institutional policies, collaborator contributions, open-source licenses and background IP. A public repository or manuscript does not prove that its author can assign exclusive rights to a new company.

Potential patent claims concern specific validated interventions, defined combinations, delivery or treatment schedules and experimentally supported applications. Broad claims to AI-guided rejuvenation face substantial prior art. A counsel-led freedom-to-operate assessment must address relevant reprogramming and delivery rights. Software patents or secrecy cannot substitute for clean biological rights.

The proposed governance structure separates experiment proposal from result acceptance. A board or designated financing committee approves major scope changes and releases later spending against evidence. Independent reviewers evaluate key results, conflicts and safety findings. Negative evidence must be available to investors and cannot be deleted because it weakens a financing narrative.

A term sheet should follow completion of the ownership register, cap table, founder commitments and material conflict disclosures. Diligence should resolve these items explicitly rather than infer them from the three author names.

Principal risks and decisions under failure

The program can fail scientifically, operationally or commercially. Each major risk has a test and a consequence. A research platform does not become investable merely by maintaining activity after a hypothesis has failed.

Risk	Evidence that would expose it	Required response
No causal rejuvenation mechanism	Clock shifts without independent function or rescue	Reject the claim and stop the candidate
Context-specific effects	Failure in new donors, laboratories or disease contexts	Narrow the indication and disclose limits
Inadequate therapeutic window	Useful function overlaps with identity loss or abnormal growth	Change modality or discontinue
No AI advantage	Expert or random selection matches confirmed hits at lower cost	Simplify the platform and reallocate compute
Inadequate statistical precision	Donor-level confidence intervals remain uninformative	Increase funded replication or narrow the claim
Expensive assays	Qualified costs exceed the planned screen envelope	Reduce target count before commitments
Weak IP or encumbered rights	Ownership review or prior-art analysis blocks intended use	Obtain rights or abandon the affected asset
No commercial demand	Buyers decline defined paid pilots despite useful science	Reassess product and financing strategy
Funding unavailable	Cash cannot support the next evidence gate	Preserve data and IP, reduce commitments or wind down

Early stop criteria include persistent inability to reproduce a reference response, absence of a measurable old-to-young functional gap in the selected assay, failure of the strongest mechanisms under orthogonal confirmation and an unacceptable identity or growth signal. A candidate may remain a useful research result while being unsuitable as a therapeutic asset.

If the computational platform is useful but the first biological hypothesis fails, a restricted research partnership may be possible. This is an alternative requiring a paying buyer and clean rights; it is not an assured recovery of investor capital. A failed therapeutic program may have little residual commercial value.

The $8.0 million plan buys a defined body of evidence and a decision. It cannot guarantee a target, a drug candidate, follow-on financing or an exit. Investors should assess whether the potential value of a validated, controllable mechanism justifies that uncertainty and whether the team can stop efficiently when the evidence is unfavorable.

Investor diligence questions on the science

Why expect a fundamental mechanism rather than a collection of markers

Reprogramming and developmental studies support the possibility of coordinated state changes. The experiment plan asks whether specific regulators cause independent functional recovery. It retains distributed-network and artifact explanations. The mechanism is a hypothesis to discover, not a premise to sell. [2–6]

What would count as genuine rejuvenation

A prespecified functional improvement toward an independent youthful reference, acceptable identity and integrity, persistence after withdrawal and replication in new donors. The exact tissue and observation window accompany every claim. A low epigenetic age alone fails this definition.

Why use skin as the first context

Accessible human cells and reconstructed tissue permit staged measurement of identity, repair and delivery. The choice is provisional and does not imply that skin captures systemic aging. A chronic-wound product would require evidence beyond fibroblast rejuvenation, including disease-relevant tissue and clinical outcomes.

Can an AI model identify a mechanism without new experiments

Observational data can rank hypotheses and reveal contradictions. Unobserved interventions and confounded relationships can remain indistinguishable regardless of model size. Targeted perturbations and appropriate controls are necessary to test causality. The seed allocation is therefore weighted toward people and experiments. [1,8,11–13]

How will cancer risk be addressed

Identity, growth and integrity are independent advancement constraints. The lead's modality determines later safety studies and surveillance. Negative short-term assays cannot establish lifetime safety, and a successful clock response cannot compensate for an abnormal-growth signal.

What if complete resetting is biologically impossible

A limited, reproducible intervention may still be a useful disease asset. The company will state the scope of its results and avoid financing an undefined claim of universal reset. If no bounded functional benefit emerges, the therapeutic thesis fails.

How will we know the AI adds value

The prospective screen compares model selection, expert nomination and random eligible candidates at equal budgets. Confirmation is blinded and independent of the selection arm. The decision metric is confirmed mechanisms per dollar, together with effect quality and reproducibility. A model that wins only a retrospective prediction metric has not met this standard.

Investor diligence questions on the business

What is the moat if models are widely available

Specific validated intervention rights, longitudinal human-cell perturbation data, independent replication and the ability to predict failure across contexts. These assets must be created. Access to an off-the-shelf model or an agent framework is not a sufficient moat.

How much capital is needed after this financing

The proposed $8.0 million supports discovery and lead feasibility for 24 months. Modality selection determines the next budget. IND-enabling studies, manufacturing, clinical trials and long-term follow-up may require substantially more capital. No reliable clinical-development total can be quoted before a lead and indication are selected.

What revenue can investors rely on

The base cash plan assumes zero commercial receipts. Two illustrative $0.5 million pilots are a commercial objective, not booked revenue. Royalties, milestones and acquisitions are excluded from the funding case because no asset or contract supports them.

How can this compete with well-funded rejuvenation companies

By producing a credible mechanism package in a narrow context at a controlled cost and establishing a specific partner opportunity. Incumbents have advantages in people, data, delivery and capital. Competitive review and partner-conflict checks must be repeated as the program develops. [18–21]

What will be available in the data room

The current research framework and archived pilot material, followed by the ownership register, founder commitments, assay qualification, raw and processed data, prespecified analyses, failed experiments, budgets, vendor scopes and independent replication. Each item will be labeled available, pending or not applicable. Missing evidence remains an explicit diligence issue.

What can prevent an investment today

Unresolved background IP, lack of experienced experimental leadership, absence of a reproducible functional assay, unclear founder commitments or an unaffordable laboratory plan. These issues have concrete remedies or a decision to stop. Authorship and ambitious goals do not resolve them.

What decision is requested from a potential investor

Assess the scientific premise, help assemble the experimental and translational team, and discuss a milestone-financed seed program after reviewing rights and assay feasibility. Specific investment structure, valuation, eligibility and terms require a separate process. We are seeking partners prepared to finance rigorous biological discovery and to act on negative results.

First ninety days and financing readiness

The first quarter must convert an attractive research idea into an executable company program. Work will be organized around four deliverables, each with a responsible leader, a budget and evidence that an investor can inspect.

Deliverable	Required contents	Acceptance condition
Formation and rights	Entity plan, founder commitments, cap table, employment conflicts, code and data rights	Authority and ownership documented
Assay qualification	Source cells, identity, reference gap, variance, controls, measurement timing and pilot costs	Reproducible functional distinction with adequate precision
Computational replay	Archived data, manifests, exact code, baseline outputs and evidence labels	Independent reproduction of the limited pilot claims
Execution plan	Experimental lead, laboratory quotes, donor procurement, analysis specification and stop rules	Scope fits the financing and confirmation is adequately powered

The discovery screen will not begin at full scale until its readouts can distinguish meaningful restoration from toxicity, selective survival and proliferation. The model benchmark will be locked before the prospective candidate comparison. A small feasibility study can change the planned assay or sample allocation, but the revised hypothesis and analysis must be recorded before confirmation begins.

By day 90, the company should have a board-reviewable release decision for the next six months. If rights, team or assay requirements remain unresolved, the remaining financing should be preserved rather than deployed into a larger screen. Investors will receive the data that support the decision, including unsuitable assays and failed reference conditions.

Financing discussions should begin with a nonconfidential summary and this white paper. Detailed target identities, sequences, unpublished combinations and enabling procedures belong in a controlled diligence process after an IP and conflict review. The outreach narrative should state the computational evidence accurately and describe biological validation as the financed work.

Our long-term ambition is to understand and control the mechanisms that reset cellular biological age. The immediate financing proposition is narrower and measurable: establish whether AI-guided discovery can produce a reproducible, controllable human-cell rejuvenation mechanism with independent functional evidence and a plausible therapeutic path. Progress toward the larger goal will follow the evidence generated by this program.

Scientific references

Sources accessed 19 October 2026. Company announcements describe reported development activity and are not independent efficacy validation. The prior manuscript is the project's current computational starting point.

[1] Xu D. Identifying the regulatory mechanisms of cellular age resetting with artificial intelligence. Research framework with public-data and synthetic computational pilots. 13 October 2026. Cellular_Age_Resetting_AI.pdf. Public pilot analyzed GSE165177; no new biological experiments.

[2] Kerepesi C et al. Epigenetic clocks reveal a rejuvenation event during embryogenesis followed by aging. Science Advances. 2021;7:eabg6082. https://doi.org/10.1126/sciadv.abg6082

[3] Kerepesi C and Gladyshev VN. Intersection clock reveals a rejuvenation event during human embryogenesis. Aging Cell. 2023;22:e13922. https://doi.org/10.1111/acel.13922

[4] Gill D et al. Multi-omic rejuvenation of human cells by maturation phase transient reprogramming. eLife. 2022;11:e71624. https://doi.org/10.7554/eLife.71624

[5] Lu Y et al. Reprogramming to recover youthful epigenetic information and restore vision. Nature. 2020;588:124–129. https://doi.org/10.1038/s41586-020-2975-4

[6] López-Otín C et al. Hallmarks of aging An expanding universe. Cell. 2023;186:243–278. https://doi.org/10.1016/j.cell.2022.11.001

[7] NCBI Gene Expression Omnibus. GSE165180 and subseries GSE165177 and GSE165179. https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE165180

[8] Roohani Y et al. Predicting transcriptional outcomes of novel multigene perturbations with GEARS. Nature Biotechnology. https://doi.org/10.1038/s41587-023-01905-6

[9] Cui H et al. scGPT Toward building a foundation model for single-cell multi-omics using generative AI. Nature Methods. 2024. https://doi.org/10.1038/s41592-024-02201-0

[10] Avsec Ž et al. Advancing regulatory variant effect prediction with AlphaGenome. Nature. 2026;649:1206–1218. https://doi.org/10.1038/s41586-025-10014-0

[11] Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines. Nature Methods. 2025. https://doi.org/10.1038/s41592-025-02772-6

[12] Systema A framework for evaluating genetic perturbation response prediction beyond systematic variation. Nature Biotechnology. 2025. https://doi.org/10.1038/s41587-025-02777-8

Development and commercial references

[13] Deep learning perturbation models can outperform baselines on calibrated metrics. Nature Biotechnology. 2026. https://doi.org/10.1038/s41587-026-03307-w

[14] Abramson J et al. Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature. 2024;630:493–500. https://doi.org/10.1038/s41586-024-07487-w

[15] FDA. Surrogate Endpoint Resources for Drug and Biologic Development. https://www.fda.gov/drugs/development-resources/surrogate-endpoint-resources-drug-and-biologic-development

[16] FDA. About Biomarkers and Qualification. https://www.fda.gov/drugs/biomarker-qualification-program/about-biomarkers-and-qualification

[17] FDA. Long Term Follow-up After Administration of Human Gene Therapy Products. Guidance for Industry. January 2020. https://www.fda.gov/regulatory-information/search-fda-guidance-documents/long-term-follow-after-administration-human-gene-therapy-products

[18] Life Biosciences. First-in-human interim data from ongoing Phase 1 ER-100 optic-neuropathy study. 8 October 2026. https://www.lifebiosciences.com/life-biosciences-announces-first-in-human-data-from-ongoing-phase-1-trial-evaluating-er-100-in-optic-neuropathies/

[19] NewLimit. NewLimit raises $435M led by Founders Fund to bring longevity medicines to human trials. 2 June 2026. https://blog.newlimit.com/p/newlimit-raises-435m-led-by-founders

[20] Altos Labs. Company mission and research focus. https://www.altoslabs.com/

[21] Retro Biosciences. Company research programs. https://www.retro.bio/

Interpretation and planning basis

All experimental volumes, hiring costs, financing needs, pilot prices, buyer counts and timing in this plan are proposed assumptions. They require assay qualification, quotations, commercial discussions and formal governance. Literature findings and competitor announcements are attributed above. Neither those findings nor the earlier computational pilot establish that this venture has discovered an anti-aging treatment.

Contact

David Xu, PhD

exutechnology@gmail.com